

❑ **Describe the postulates of Bohr's Atomic Model with its limitations.**

Postulates:

1. **Quantized Energy Levels:** Electrons orbit the nucleus in specific, discrete energy levels or shells. These energy levels are quantized, meaning that electrons can only occupy certain energy states.
2. **Stationary States:** While in these stationary states, electrons do not radiate energy. They remain stable until they transition to a different energy level.
3. **Energy Transitions:** Electrons can transition between energy levels by absorbing or emitting photons of light. The energy of the photon corresponds to the difference in energy between the two levels.
4. **Angular Momentum:** The angular momentum of an electron in a particular orbit is quantized. It can only have certain values, determined by the quantum number n .

Limitations:

1. **Fine Structure and Zeeman Effect:** Bohr's model could not explain the fine structure of spectral lines, which arises from the interaction between the electron's spin and its orbital motion. It also couldn't explain the Zeeman effect, the splitting of spectral lines in a magnetic field.
2. **Relativistic Effects:** At high speeds, relativistic effects become significant, and Bohr's model fails to accurately describe the behavior of electrons.

3. **Quantum Mechanics:** Bohr's model, while a significant step forward, was ultimately superseded by quantum mechanics, which provides a more accurate and complete description of the atom.

In summary, Bohr's atomic model provided a valuable framework for understanding the structure of the atom, but it had limitations that were addressed by the development of quantum mechanics.

❑ What is the difference between orbit and orbital. Write the electronic configuration and mention the position of the following elements in the periodic table: a. Cr (24), K (19), Br (35).

The terms "orbit" and "orbital" are often used interchangeably in chemistry, but they have distinct meanings:

- **Orbit:** In Bohr's model of the atom, an orbit is a fixed path around the nucleus where an electron can travel. It has a specific energy level and shape.
- **Orbital:** In modern quantum mechanics, an orbital is a region of space where there is a high probability of finding an electron. Orbitals are more complex than Bohr's orbits and can have different shapes (s, p, d, f).

Electronic Configurations:

- **Cr (24):** $[\text{Ar}] 3d^5 4s^1$

- **K (19):** $[\text{Ar}] 4s^1$
- **Br (35):** $[\text{Ar}] 3d^{10} 4s^2 4p^5$

Positions in the Periodic Table:

- **Cr (24):** Period 4, Group 6 (Transition metals)
- **K (19):** Period 4, Group 1 (Alkali metals)
- **Br (35):** Period 4, Group 17 (Halogens).

☐ **State Mendeleev's periodic law. Discuss the s,p,d and f-block elements with examples.**

Mendeleev's periodic law states that the properties of elements are a periodic function of their atomic masses. In other words, when elements are arranged in order of increasing atomic mass, their properties repeat in a regular pattern.

s, p, d, and f-block elements:

The periodic table is divided into four blocks based on the type of orbital that the outermost electrons occupy:

- **s-block elements:** These elements have their outermost electrons in the s orbital. They are located in the first two groups of the periodic table. Examples of s-block elements include hydrogen, lithium, sodium, potassium, beryllium, magnesium, calcium, and strontium.
- **p-block elements:** These elements have their outermost electrons in the p orbital. They are located in groups 13 to 18 of

the periodic table. Examples of p-block elements include boron, carbon, nitrogen, oxygen, fluorine, neon, aluminum, silicon, phosphorus, sulfur, chlorine, and argon.

- **d-block elements:** These elements have their outermost electrons in the d orbital. They are located in groups 3 to 12 of the periodic table and are also known as transition metals. Examples of d-block elements include scandium, titanium, vanadium, chromium, manganese, iron, cobalt, nickel, copper, zinc, and many others.
- **f-block elements:** These elements have their outermost electrons in the f orbital. They are located in the two rows below the main body of the periodic table and are also known as lanthanides and actinides. Examples of f-block elements include lanthanum, cerium, praseodymium, neodymium, and uranium.

The properties of elements are related to the configuration of their outermost electrons. For example, elements in the same group of the periodic table have similar chemical properties because they have the same number of valence electrons (outermost electrons).

☐ **Briefly discuss about the Molecular Orbital Theory (MOT).**

The Molecular Orbital Theory (MOT) is a quantum mechanical theory that describes the behavior of electrons in molecules. It is based on

the idea that when atoms combine to form a molecule, their atomic orbitals overlap to form molecular orbitals. These molecular orbitals are delocalized over the entire molecule, meaning that the electrons are not confined to any single atom.

MOT is used to explain the bonding, structure, and properties of molecules. It can also be used to predict the reactivity and stability of molecules.

One of the key concepts in MOT is the idea of bonding and antibonding orbitals. Bonding orbitals are formed when atomic orbitals combine constructively, resulting in an increase in electron density between the atoms. Antibonding orbitals are formed when atomic orbitals combine destructively, resulting in a decrease in electron density between the atoms.

The number and type of molecular orbitals formed in a molecule depend on the number and types of atomic orbitals that are involved in the bonding. For example, two hydrogen atoms can combine to form a sigma bonding orbital and a sigma antibonding orbital.

MOT is a powerful tool for understanding the electronic structure of molecules. It has been used to explain a wide variety of phenomena, including chemical bonding, spectroscopy, and reactivity.

❑ What do you mean by Lanthanides and Actinides? Discuss briefly about general properties of Lanthanides.

Lanthanides and actinides are two groups of elements in the periodic table that are often referred to as the **f-block elements**. They are located below the main body of the periodic table.

Lanthanides:

- The 14 elements from lanthanum (La) to lutetium (Lu) are collectively known as lanthanides.
- They are also referred to as rare earth elements, although they are not actually rare.
- Lanthanides have similar physical and chemical properties due to their similar electronic configuration.
- They are typically hard, silvery-white metals with high melting and boiling points.
- Lanthanides are used in a variety of applications, including electronics, metallurgy, and medicine.

Actinides:

- The 15 elements from actinium (Ac) to lawrencium (Lr) are collectively known as actinides.
- They are all radioactive elements.
- Actinides have a wide range of properties, depending on their specific element.
- Many actinides are used in nuclear technology and research.

General Properties of Lanthanides:

- **Similar chemical properties:** Lanthanides have similar chemical properties due to their similar electronic

configuration. This makes it difficult to separate them from each other.

- **High melting and boiling points:** Lanthanides are generally hard, silvery-white metals with high melting and boiling points.
- **Magnetic properties:** Many lanthanides exhibit magnetic properties, such as paramagnetism and ferromagnetism.
- **Color:** Lanthanides often have distinctive colors due to the transitions of electrons between different energy levels.
- **Reactivity:** Lanthanides are generally reactive metals, especially when exposed to air or water.

Overall, lanthanides are a group of elements with unique properties that have a wide range of applications in various fields.